

# LOW STYRENE SBR PATENT LANDSCAPE: EMULSION, SOLUTION, AND MODIFIED BINDER ROUTES

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## 1. ABSTRACT

This report synthesizes seven primary patents relevant to low styrene styrene-butadiene rubber (SBR) and related styrene-butadiene copolymers. The target is defined as SBR materials with styrene content below conventional SBR levels, typically butadiene-rich rubber/elastomer grades used in tires, tread compounds, and functional binders. Across the selected patents, the landscape spans both solution polymerization and emulsion polymerization, with several documents emphasizing downstream functionalization, oil extension, or binder architecture rather than only base copolymer synthesis. The solution-polymerized disclosures (e.g., tapered or modified SBR in hydrocarbon solvent) use organolithium initiation, hydrocarbon solvents, and post-polymerization quench/coagulation or solvent removal, enabling control over molecular weight and sequence distribution. The emulsion-polymerized disclosures rely on water as the polymerization medium, typically with rosin/fatty-acid soaps, redox initiator systems, mercaptan molecular-weight control, and in some cases reactive emulsifiers or epoxy-acrylate monomers to introduce polarity or silica affinity. Several patents also include post-polymerization water steps such as coagulation, washing, stripping, or extraction, which are important for isolating the rubber or binder. A recurring formulation trend is the use of these low-styrene SBRs in silica- or carbon-black-filled tire tread compounds, where the patents report improved wet grip, wear resistance, processability, or reduced oil demand. Another major trend is tailoring the styrene fraction and microstructure: one patent claims a tapered SBR with 20-30 wt% styrene and 70-80 wt% butadiene, while others disclose emulsion SBRs with styrene around 26-27 wt% or 15-50 wt% depending on functionalization strategy. Overall, the selected patents show that low-styrene SBR development is driven by balancing rolling resistance, wear, and processability through controlled monomer feed, polymerization medium, and post-treatment/compounding design.

## 2. METHODOLOGY

### PRIMARY PATENT EVIDENCE

#### Patent 1

- Patent Number: US11591428B2
- Patent Title: Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ...

- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: The patent discloses preparation of a tapered styrene-butadiene copolymer by solution polymerization of styrene and 1,3-butadiene in hydrocarbon solvent (hexane) using an organolithium initiator and sodium dodecylbenzenesulfonate. The cement is later coagulated in isopropanol containing BHT and drum dried to isolate the polymer. - Hexane: 0.18 kg — Example 1 - Styrene: 0.36 kg of 31.8 wt % in hexane — Example 1 - 1,3-Butadiene: 1.68 kg of 20.2 wt % in hexane — Example 1 - SDBS: 2.27 mL of 0.5 M in ethylbenzene — Example 1 - n-Butyl lithium: 6.05 mL of 1.60 M in hexane — Example 1 - Reaction temperature: 60 °C — Example 1 - Peak batch temperature: 69 °C — Example 1 - Polymerization time: 24 minutes + additional 2 hours — Example 1 - Styrene content: 26.6 % — Example 1 - 1,4-butadiene incorporation: 64 % — Example 1 - Mn: 52 kg/mol — Example 1 - Mw: 55 kg/mol — Example 1 - PDI: 1.05 — Example 1 - Tg: -76.07 °C — Example 1 - Hexane: 0.30 kg — Example 2 - Styrene: 0.55 kg of 32.9 wt % in hexane — Example 2 - 1,3-Butadiene: 2.72 kg of 20 wt % in hexane — Example 2 - SDBS: 3.63 mL of 0.5 M in ethylbenzene — Example 2 - n-Butyl lithium: 9.07 mL of 1.60 M in hexane — Example 2 - Reaction temperature: 66 °C — Example 2 - Peak batch temperature: 107 °C — Example 2 - Polymerization time: 20 minutes + additional 40 minutes — Example 2 - Styrene content: 25.7 % — Example 2 - 1,4-butadiene incorporation: 64 % — Example 2 - Mn: 66 kg/mol — Example 2 - Mw: 74 kg/mol — Example 2 - PDI: 1.12 — Example 2 - Tg: -78.79 °C — Example 2 - Total styrene content: 20 to 30 percent by weight — Claim 1 - Total butadiene content: 70 to 80 percent by weight — Claim 1 - Styrene sequences >3 units: greater than 30 weight percent of total styrene — Claim 1 - Styrene sequences >10 units: less than 20 weight percent of total styrene — Claim 1 - Number average molecular weight: 30,000 to 80,000 — Claim 1 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR levels: 20 to 30 percent by weight of the styrene-butadiene copolymer; examples at 26.6% and 25.7% styrene

**Medium & Water Role** - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic hydrocarbon solution polymerization (hexane cement)

**styrene content below conventional SBR levels** - 20 to 30 percent by weight of the styrene-butadiene copolymer; examples at 26.6% and 25.7% styrene

**Relevant Experimental Evidence** - Example 1: A tapered SBR cement was prepared in hexane and isolated by isopropanol coagulation; the product showed 26.6% styrene and low PDI. - Example 2: A second tapered SBR was prepared under similar solution conditions with slightly higher temperature excursion and 25.7% styrene. - Claim 1: The target copolymer is defined as 20-30 wt% styrene and 70-80 wt% butadiene with controlled styrene sequence distribution and Mn

30,000-80,000.

**Technical Relevance** This patent is directly relevant because it defines a low-styrene SBR architecture and shows how solution polymerization can be used to control styrene content, sequence distribution, and molecular weight. The tapered microstructure is especially pertinent to tire-grade low-styrene SBR design.

## Patent 2

- Patent Number: US8530017B2
- Patent Title: Method for preparing styrene-butadiene copolymer using reactive emulsifier and ...
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: The patent discloses radical emulsion polymerization of styrene and 1,3-butadiene in water using a reactive emulsifier that also participates in the polymerization. The resulting SBR latex is stripped and dried, and the patent emphasizes improved silica affinity and tire tread performance. - Water: 1600 mL — Comparative Example 1 - Potassium rosinate: 22 g — Comparative Example 1 - Fatty acid potassium salt: 36 g — Comparative Example 1 - Potassium chloride: 1.5 g — Comparative Example 1 - Styrene: 290 g — Comparative Example 1 - 1,3-Butadiene: 710 g — Comparative Example 1 - EDTA: 0.5 g — Comparative Example 1 - Sodium hydrosulfite: 0.3 g — Comparative Example 1 - Ferrous sulfate: 0.1 g — Comparative Example 1 - Sodium formaldehyde sulfonate: 0.4 g — Comparative Example 1 - n-Dodecyl mercaptan: 1.0 g — Comparative Example 1 - Methane hydroperoxide: 0.6 g — Comparative Example 1 - Polymerization time: 8 hours — Comparative Example 1 - Polymerization temperature: 10 °C — Comparative Example 1 - Water: 1600 mL — Example 1 - Potassium rosinate: 11 g — Example 1 - Fatty acid potassium salt: 18 g — Example 1 - Potassium chloride: 1.5 g — Example 1 - Styrene: 260 g — Example 1 - Methylcarboxylate methacrylate: 30 g — Example 1 - 1,3-Butadiene: 710 g — Example 1 - EDTA: 0.5 g — Example 1 - Sodium hydrosulfite: 0.3 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - Sodium formaldehyde sulfonate: 0.4 g — Example 1 - n-Dodecyl mercaptan: 1.0 g — Example 1 - Methane hydroperoxide: 0.6 g — Example 1 - Polymerization time: 8 hours — Example 1 - Polymerization temperature: 10 °C — Example 1 - Weight average molecular weight: 860,000 g/mol — Example 1 - Water: 1600 mL — Example 2 - Potassium rosinate: 11 g — Example 2 - Fatty acid potassium salt: 18 g — Example 2 - Potassium chloride: 1.5 g — Example 2 - Styrene: 280 g — Example 2 - Polyoxyethylene phosphate methacrylate: 10 g — Example 2 - 1,3-Butadiene: 710 g — Example 2 - EDTA: 0.5 g — Example 2 - Sodium hydrosulfite: 0.3 g — Example 2 - Ferrous sulfate: 0.1 g — Example 2 - Sodium formaldehyde sulfonate: 0.4 g — Example 2 - n-Dodecyl mercaptan: 1.0 g — Example 2 - Methane hydroperoxide: 0.6 g — Example 2 - Polymerization time: 8 hours —

Example 2 - Polymerization temperature: 10 °C — Example 2 - Weight average molecular weight: 820,000 g/mol — Example 2 - Water: 1600 mL — Example 3 - Potassium rosinat: 11 g — Example 3 - Fatty acid potassium salt: 18 g — Example 3 - Potassium chloride: 1.5 g — Example 3 - Styrene: 260 g — Example 3 - Polyoxyethylene phosphate methacrylate: 30 g — Example 3 - 1,3-Butadiene: 710 g — Example 3 - EDTA: 0.5 g — Example 3 - Sodium hydrosulfite: 0.3 g — Example 3 - Ferrous sulfate: 0.1 g — Example 3 - Sodium formaldehyde sulfonate: 0.4 g — Example 3 - n-Dodecyl mercaptan: 1.0 g — Example 3 - Methane hydroperoxide: 0.6 g — Example 3 - Polymerization time: 8 hours — Example 3 - Polymerization temperature: 10 °C — Example 3 - Weight average molecular weight: 824,000 g/mol — Example 3 - Water: 1600 mL — Example 4 - Potassium rosinat: 11 g — Example 4 - Fatty acid potassium salt: 18 g — Example 4 - Potassium chloride: 1.5 g — Example 4 - Styrene: 260 g — Example 4 - Allyl alcohol ethoxylate: 30 g — Example 4 - 1,3-Butadiene: 710 g — Example 4 - EDTA: 0.5 g — Example 4 - Sodium hydrosulfite: 0.3 g — Example 4 - Ferrous sulfate: 0.1 g — Example 4 - Sodium formaldehyde sulfonate: 0.4 g — Example 4 - n-Dodecyl mercaptan: 1.0 g — Example 4 - Methane hydroperoxide: 0.6 g — Example 4 - Polymerization time: 8 hours — Example 4 - Polymerization temperature: 10 °C — Example 4 - Weight average molecular weight: 850,000 g/mol — Example 4 - Water: 1600 mL — Example 5 - Potassium rosinat: 11 g — Example 5 - Fatty acid potassium salt: 18 g — Example 5 - Potassium chloride: 1.5 g — Example 5 - Styrene: 260 g — Example 5 - Allyl alcohol polyethylene oxide phosphate: 30 g — Example 5 - 1,3-Butadiene: 710 g — Example 5 - EDTA: 0.5 g — Example 5 - Sodium hydrosulfite: 0.3 g — Example 5 - Ferrous sulfate: 0.1 g — Example 5 - Sodium formaldehyde sulfonate: 0.4 g — Example 5 - n-Dodecyl mercaptan: 1.0 g — Example 5 - Methane hydroperoxide: 0.6 g — Example 5 - Polymerization time: 8 hours — Example 5 - Polymerization temperature: 10 °C — Example 5 - Weight average molecular weight: 900,000 g/mol — Example 5 - Water: 1600 mL — Example 6 - Potassium rosinat: 11 g — Example 6 - Fatty acid potassium salt: 18 g — Example 6 - Potassium chloride: 1.5 g — Example 6 - Styrene: 260 g — Example 6 - Allyl polyoxyethylene glycol methacrylate: 30 g — Example 6 - 1,3-Butadiene: 710 g — Example 6 - EDTA: 0.5 g — Example 6 - Sodium hydrosulfite: 0.3 g — Example 6 - Ferrous sulfate: 0.1 g — Example 6 - Sodium formaldehyde sulfonate: 0.4 g — Example 6 - n-Dodecyl mercaptan: 1.0 g — Example 6 - Methane hydroperoxide: 0.6 g — Example 6 - Polymerization time: 8 hours — Example 6 - Polymerization temperature: 10 °C — Example 6 - Weight average molecular weight: 950,000 g/mol — Example 6 - Water: 1600 mL — Example 7 - Potassium rosinat: 11 g — Example 7 - Fatty acid potassium salt: 18 g — Example 7 - Potassium chloride: 1.5 g — Example 7 - Styrene: 260 g — Example 7 - Polyoxypropylene glycol phosphate acrylate: 30 g — Example 7 - 1,3-Butadiene: 710 g — Example 7 - EDTA: 0.5 g — Example 7 - Sodium hydrosulfite: 0.3 g — Example 7 - Ferrous sulfate: 0.1 g — Example 7 - Sodium formaldehyde sulfonate: 0.4 g — Example 7 - n-Dodecyl mercaptan: 1.0 g — Example 7 - Methane hydroperoxide: 0.6 g — Example 7 - Polymerization time: 8 hours — Example 7 - Polymerization temperature: 10 °C — Example 7 - Weight average molecular weight: 845,000 g/mol — Example 7 - Water: 8,700 g — Example 1 - Coagulation temperature: 65 °C — Example 1 - Drying temperature: 65 °C — Example 1 - Drying time: 18 hours — Example 1 - Oil in dry rubber: 37.5 phr — Example 1 - Water: 8,500 g — Example 4 - Buffering system pH: 8.7 — Example 4 - Polymerization duration: about 5 hours — Example 4 - Final latex pH: 9.2 — Example 4 - Oil in dry rubber: 37.5 phr — Example 4 - Medium & Water Role: Water is the polymerization medium for emulsion SBR latex formation, and it is also used later for coagulation, washing, and

latex handling/workup. - styrene content below conventional SBR levels: Styrene used at 260-280 g in Examples 1-7; Example 6 reports bound styrene content of 21.0 wt%

**Medium & Water Role** - Water is the polymerization medium for emulsion SBR latex formation, and it is also used later for coagulation, washing, and latex handling/workup. - Core reaction medium: Aqueous emulsion/latex polymerization

**styrene content below conventional SBR levels** - Styrene used at 260-280 g in Examples 1-7; Example 6 reports bound styrene content of 21.0 wt%

**Relevant Experimental Evidence** - Example 1: Standard SBR latex was prepared in water and then blended with extension oil, coagulated, washed, and dried to obtain oil-extended rubber. - Example 4: Functionalized SBR latex was prepared with GMA and a buffered aqueous emulsion system, then oil-extended and isolated by coagulation/washing/drying. - Example 6: Functionalized SBR with epoxidized soybean oil extension showed bound styrene content of 21.0 wt% and detailed microstructure data.

**Technical Relevance** This patent is relevant because it discloses emulsion SBR synthesis with styrene levels in the low-to-mid 20 wt% range and demonstrates functionalized variants for silica-compatible tire tread use. It also shows how water-based polymerization and downstream coagulation/washing are integrated into industrial SBR manufacture.

### Patent 3

- Patent Number: US9328176B2
- Patent Title: Functional styrene-butadiene copolymer
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: The patent discloses emulsion polymerization of styrene, 1,3-butadiene, and an epoxy acrylate monomer in water, followed by ring-opening of the epoxy group under acidic or alkaline conditions to form a functionalized SBR. The product is then used in silica-filled tire tread compounds. - Water: 1,500 mL — Comparative Example 1 - Sodium rosinat: 25 g — Comparative Example 1 - Sodium salt of fatty acid: 35 g — Comparative Example 1 - Styrene: 400 g — Comparative Example 1 - 1,3-Butadiene: 600 g — Comparative Example 1 - EDTA: 0.6 g — Comparative Example 1 - Ferrous sulfate: 0.1 g — Comparative Example 1 - n-Dodecylmercaptan: 1.5 g — Comparative Example 1 - p-Menthane hydroperoxide: 0.6 g — Comparative Example 1 - Polymerization temperature: 10 °C — Comparative Example 1 - Polymerization time: 24 hours — Comparative Example 1 - Average molecular weight: 843,000 g/mol — Comparative Example 1 - Water: 1,500 mL — Comparative

Example 2 - Sodium rosinat: 25 g — Comparative Example 2 - Sodium salt of fatty acid: 35 g — Comparative Example 2 - Styrene: 380 g — Comparative Example 2 - 1,3-Butadiene: 570 g — Comparative Example 2 - Methyl methacrylic acid: 50 g — Comparative Example 2 - EDTA: 0.6 g — Comparative Example 2 - Ferrous sulfate: 0.1 g — Comparative Example 2 - n-Dodecylmercaptan: 1.5 g — Comparative Example 2 - p-Menthane hydroperoxide: 0.6 g — Comparative Example 2 - Polymerization temperature: 10 °C — Comparative Example 2 - Polymerization time: 24 hours — Comparative Example 2 - Average molecular weight: 795,000 g/mol — Comparative Example 2 - Water: 1,500 mL — Example 1 - Sodium rosinat: 25 g — Example 1 - Sodium salt of fatty acid: 35 g — Example 1 - Styrene: 380 g — Example 1 - Glycidyl methyl methacrylate: 50 g — Example 1 - 1,3-Butadiene: 570 g — Example 1 - EDTA: 0.6 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - n-Dodecylmercaptan: 1.5 g — Example 1 - p-Menthane hydroperoxide: 0.6 g — Example 1 - Polymerization temperature: 10 °C — Example 1 - Polymerization time: 24 hours — Example 1 - Sulfuric acid (20% aqueous solution): 100 g — Example 1 - Average molecular weight: 908,000 g/mol — Example 1 - o: 0.38 — Example 1 - p: 0.57 — Example 1 - q: 0.05 — Example 1 - Water: 1,500 mL — Example 4 - Sodium rosinat: 25 g — Example 4 - Sodium salt of fatty acid: 35 g — Example 4 - Styrene: 380 g — Example 4 - 1,3-Butadiene: 570 g — Example 4 - EDTA: 0.6 g — Example 4 - Ferrous sulfate: 0.1 g — Example 4 - n-Dodecylmercaptan: 1.5 g — Example 4 - p-Menthane hydroperoxide: 0.6 g — Example 4 - Polymerization temperature: 10 °C — Example 4 - Polymerization time: 24 hours — Example 4 - NaOH (20% aqueous solution): 100 g — Example 4 - Average molecular weight: 740,000 g/mol — Example 4 - o: 0.38 — Example 4 - p: 0.57 — Example 4 - q: 0.05 — Example 4 - Water: 1,500 mL — Example 5 - Sodium rosinat: 25 g — Example 5 - Sodium salt of fatty acid: 35 g — Example 5 - Styrene: 380 g — Example 5 - 1,3-Butadiene: 570 g — Example 5 - EDTA: 0.6 g — Example 5 - Ferrous sulfate: 0.1 g — Example 5 - n-Dodecylmercaptan: 1.5 g — Example 5 - p-Menthane hydroperoxide: 0.6 g — Example 5 - Polymerization temperature: 10 °C — Example 5 - Polymerization time: 24 hours — Example 5 - KOH (20% aqueous solution): 100 g — Example 5 - Average molecular weight: 791,000 g/mol — Example 5 - o: 0.38 — Example 5 - p: 0.57 — Example 5 - q: 0.05 — Example 5 - Medium & Water Role: Water is the emulsion polymerization medium, and aqueous acid/base solutions are also used in the ring-opening post-treatment of the polymer. - styrene content below conventional SBR levels: Styrene used at 380 g in Example 1 and 260-380 g in comparative/examples; no explicit wt% styrene for the target copolymer is given in the extracted evidence

**Medium & Water Role** - Water is the emulsion polymerization medium, and aqueous acid/base solutions are also used in the ring-opening post-treatment of the polymer. - Core reaction medium: Aqueous emulsion polymerization

**styrene content below conventional SBR levels** - Styrene used at 380 g in Example 1 and 260-380 g in comparative/examples; no explicit wt% styrene for the target copolymer is given in the extracted evidence

**Relevant Experimental Evidence** - Example 1: GMA-containing SBR was prepared in water and then ring-opened with 20% sulfuric acid to give a hydroxyl-functional copolymer. - Example 4: The same emulsion/ring-opening route was repeated using 20% NaOH as the ring-opening agent. -

Example 5: The same route was repeated using 20% KOH as the ring-opening agent, giving a different molecular weight product.

**Technical Relevance** This patent is relevant because it shows a low-to-moderate styrene emulsion SBR platform that is further functionalized for silica compatibility and tire tread performance. The disclosed styrene/butadiene feed levels and aqueous emulsion process are directly applicable to low-styrene SBR synthesis.

#### Patent 4

- Patent Number: EP2986647B1
- Patent Title: Oil extended functionalized styrene-butadiene copolymer
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: The patent discloses preparation of an oil-extended functionalized SBR latex by emulsion polymerization in water, using a potassium salt of fatty and rosin acids as emulsifier and a reactive monomer such as glycidyl methacrylate. The latex is blended with extension oil emulsion, coagulated with steam and sulfuric acid, washed with demineralized water, and dried to obtain the rubber. - Water: 8,700 g — Example 1 - Potassium salt of fatty and rosin acids: 2,330 g — Example 1 - 1,3-Butadiene: 3,830 g — Example 1 - Styrene: 1,486 g — Example 1 - Pinane hydroperoxide: 2.5 g — Example 1 - Activator batch with water, ferrous sulfate, EDTA and sodium formaldehyde sulfoxylate: 390 g — Example 1 - t-Dodecyl mercaptan: 10 g — Example 1 - Polymerization temperature: 7 °C — Example 1 - Conversion: 60 % — Example 1 - Polymerization duration: about 7 hours — Example 1 - Oil in dry rubber: 37.5 phr — Example 1 - Coagulation temperature: 65 °C — Example 1 - Drying temperature: 65 °C — Example 1 - Drying time: 18 hours — Example 1 - Water: 8,500 g — Example 4 - Potassium salt of fatty and rosin acids: 2,330 g — Example 4 - Phosphoric acid 85 %: 2.1 g — Example 4 - Acetic acid 99 %: 5.4 g — Example 4 - Potassium hydroxide 50 %: 15.3 g — Example 4 - Water in buffering system: 200.0 g — Example 4 - pH: 8.7 — Example 4 - 1,3-Butadiene: 3,830 g — Example 4 - Styrene: 1,242 g — Example 4 - Blend of GMA/styrene: 106/133 g — Example 4 - Pinane hydroperoxide: 2.5 g — Example 4 - Activator batch with water, ferrous sulfate, EDTA and sodium formaldehyde sulfoxylate: 390 g — Example 4 - t-Dodecyl mercaptan: 10 g — Example 4 - Polymerization temperature: 7 °C — Example 4 - Conversion: 60 % — Example 4 - Polymerization duration: about 5 hours — Example 4 - Final latex pH: 9.2 — Example 4 - Oil in dry rubber: 37.5 phr — Example 4 - Bound styrene content: 21.0 %wt — Example 6 - cis-1.4: 9.5 %wt — Example 6 - trans-1.4: 76.3 %wt — Example 6 - vinyl 1.2: 14.2 %wt — Example 6 - Mooney viscosity (ML 1+4, 100°C): 45.8 — Example 6 - Average molecular weight (Mw): 356,000 g/mol — Example 6 - Styrene repeating units: 15 to 50 wt.% — Claim 1 - Butadiene repeating units: 42 to 80 wt.% — Claim 1 - Functional monomer units: 5 to 43 wt.% —

Claim 1 - Functional monomer units: 0.3 to 10 wt.% — Claim 3 - Mooney viscosity (ML 1+4 at 100°C): 35 to 65 — Claim 4 - Average molecular weight (Mw): 100,000 to 2,200,000 g/mol — Claim 4 - Medium & Water Role: Water is the core emulsion polymerization medium, and it is also used later in latex blending, coagulation with steam and sulfuric acid, washing, and drying workup. - styrene content below conventional SBR levels: Bound styrene content: 21.0 %wt

**Medium & Water Role** - Water is the core emulsion polymerization medium, and it is also used later in latex blending, coagulation with steam and sulfuric acid, washing, and drying workup. - Core reaction medium: Aqueous emulsion/latex polymerization

**styrene content below conventional SBR levels** - Bound styrene content: 21.0 %wt

**Relevant Experimental Evidence** - Example 1: Standard SBR latex was made in water, then oil-extended, coagulated with steam and sulfuric acid, washed, and dried. - Example 4: Functionalized SBR latex was prepared with GMA/styrene feed and buffered aqueous emulsion conditions, then oil-extended and isolated. - Example 6: The functionalized oil-extended SBR showed 21.0 wt% bound styrene and detailed microstructure/Mooney data.

**Technical Relevance** This patent is highly relevant because it discloses an emulsion SBR platform with styrene content in the low-20 wt% range and functionalization for silica-compatible tire tread use. It also provides a clear industrial route from aqueous latex to isolated rubber via coagulation, washing, and drying.

## Patent 5

- Patent Number: US12037432B2
- Patent Title: Method for preparing modified and conjugated diene-based polymer and modified ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in n-hexane using n-butyllithium, followed by terminal modification with an alkoxysilane-based modifier and a subsequent condensation stabilization step with a modification stabilizer. The polymer is then quenched with ethanol and isolated by hot-water solvent removal and roll drying. - Styrene: 270 g — Example 1 - 1,3-Butadiene: 710 g — Example 1 - n-Hexane: 5,000 g — Example 1 - 2,2-di(2-tetrahydrofuryl)propane: 1.3 g — Example 1 - Internal temperature: 40 °C — Example 1 - n-Butyllithium: 4 mmol — Example 1 - Terminal capping butadiene: 20 g — Example 1 - Modifier: 4 mmol of a compound of Formula 5a — Example 1 - Ethanol: 0.03 parts by weight based on monomers — Example 1 - BHT solution: 5 ml

of hexane solution containing 0.3 wt % BHT — Example 1 - Condensation reaction time: 15 minutes — Example 1 - Hot water: 100 °C — Example 1 - Styrene: 270 g — Comparative Example 1 - 1,3-Butadiene: 710 g — Comparative Example 1 - n-Hexane: 5,000 g — Comparative Example 1 - 2,2-di(2-oxolanyl)propane: 1.3 g — Comparative Example 1 - Internal temperature: 40 °C — Comparative Example 1 - n-Butyllithium: 4 mmol — Comparative Example 1 - Terminal capping butadiene: 20 g — Comparative Example 1 - Modifier: 4 mmol of dimethylaminomethoxysilane — Comparative Example 1 - Ethanol: 0.03 parts by weight based on monomers — Comparative Example 1 - BHT solution: 5 ml of hexane solution containing 0.3 wt % BHT — Comparative Example 1 - Hot water: 100 °C — Comparative Example 1 - Mooney viscosity (MV): 67 — Example 1 - Mooney viscosity (MV): 69 — Example 2 - Mooney viscosity (MV): 73 — Example 3 - Mooney viscosity (MV): 97 — Comparative Example 1 - Styrene content: 27.0 wt % based on total polymer amount — Example 1 - Vinyl content: 42.9 wt % based on total polymer amount — Example 1 - Mn:  $12.6 \times 10^4$  g/mol — Example 1 - Mw:  $13.6 \times 10^4$  g/mol — Example 1 - PDI: 1.08 — Example 1 - Styrene content: 26.8 wt % based on total polymer amount — Example 2 - Vinyl content: 43.1 wt % based on total polymer amount — Example 2 - Mn:  $13.2 \times 10^4$  g/mol — Example 2 - Mw:  $14.8 \times 10^4$  g/mol — Example 2 - PDI: 1.12 — Example 2 - Styrene content: 26.9 wt % based on total polymer amount — Example 3 - Vinyl content: 42.8 wt % based on total polymer amount — Example 3 - Mn:  $13.0 \times 10^4$  g/mol — Example 3 - Mw:  $15.1 \times 10^4$  g/mol — Example 3 - PDI: 1.16 — Example 3 - Styrene content: 27.1 wt % based on total polymer amount — Comparative Example 1 - Vinyl content: 42.9 wt % based on total polymer amount — Comparative Example 1 - Mn:  $16.9 \times 10^4$  g/mol — Comparative Example 1 - Mw:  $20.4 \times 10^4$  g/mol — Comparative Example 1 - PDI: 1.21 — Comparative Example 1 - Medium & Water Role: The polymerization itself is carried out in hydrocarbon solvent, but water is used later as hot water for solvent removal during isolation of the modified SBR. - styrene content below conventional SBR levels: 27.0 wt % based on total polymer amount

**Medium & Water Role** - The polymerization itself is carried out in hydrocarbon solvent, but water is used later as hot water for solvent removal during isolation of the modified SBR. - Core reaction medium: Organic hydrocarbon solution polymerization (n-hexane)

**styrene content below conventional SBR levels** - 27.0 wt % based on total polymer amount

**Relevant Experimental Evidence** - Example 1: A modified SBR was prepared in n-hexane, quenched with ethanol, stabilized by condensation, and isolated by hot-water solvent removal. - Example 2: The same route with a different stabilizer gave a similar low-styrene modified SBR with slightly higher Mooney viscosity. - Example 3: Another stabilizer variant again produced a low-styrene modified SBR with comparable styrene and vinyl contents.

**Technical Relevance** This patent is relevant because it provides a solution-polymerized SBR route with explicit styrene content around 27 wt%, which is within low-styrene rubber-grade territory. The process also addresses post-polymerization stabilization and solvent removal, both important for industrial SBR manufacture.

**Patent 6**

- Patent Number: US5902856A
- Patent Title: Rubber composition of two diene rubbers
- Assignee: Nippon Zeon Co., Ltd.
- Jurisdiction: US
- Publication Year: 1999
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: The patent discloses blending a terminal-modified diene polymer solution with a high-molecular diene polymer solution in solution state, followed by steam stripping, water squeezing, and vacuum drying to obtain a diene rubber. The resulting rubber is then compounded and vulcanized for property evaluation. - St/V: 21/63 — Table 1 - M1: 50000 — Table 1 - St/V: 21/66 — Table 1 - M1: 150000 — Table 1 - St/V: 21/65 — Table 1 - M1: 210000 — Table 1 - St/V: 21/65 — Table 1 - M1: 250000 — Table 1 - St/V: 21/65 — Table 1 - M1: 300000 — Table 1 - St/V: 21/65 — Table 1 - M1: 360000 — Table 1 - St/V: 21/64 — Table 1 - M1: 650000 — Table 1 - M2: 1200000 — Table 1 - M2: 900000 — Table 1 - M2: 600000 — Table 1 - St/V: 15/28 — Table 2 - Tg: -60 °C — Table 2 - M1: 220000 — Table 2 - M2: 1200000 — Table 2 - St/V: 23/24 — Table 3 - Tg: -50 °C — Table 3 - M1: 230000 — Table 3 - M2: 1100000 — Table 3 - St/V: 21/65 — Table 5 - M1: 250000 — Table 5 - M2: 1200000 — Table 5 - St/V: 21/65 — Table 6 - M1: 250000 — Table 6 - M2: 1200000 — Table 6 - St/V: 21/65 — Table 7 - M1: 250000 — Table 7 - M2: 1200000 — Table 7 - St/V: 21/65 — Table 8 - M1: 250000 — Table 8 - M2: 1200000 — Table 8 - St/V: 23/34 — Table 10 - Tg: -50 °C — Table 10 - M1: 230000 — Table 10 - M2: 1150000 — Table 10 - Steam stripping temperature: 65 °C — Example 1 - Vacuum drying temperature: 60 °C — Example 1 - Vacuum drying time: 24 hours — Example 1 - Press vulcanization temperature: 160 °C — Example 1 - Press vulcanization time: 25 minutes — Example 1 - Medium & Water Role: The rubber components are handled in solution, but water appears later during steam stripping/coagulation and water squeezing before drying. - styrene content below conventional SBR levels: St/V values include 21/63, 21/64, 21/65, 21/66 and 23/34; no explicit wt% styrene for the requested target material is disclosed

**Medium & Water Role** - The rubber components are handled in solution, but water appears later during steam stripping/coagulation and water squeezing before drying. - Core reaction medium: Solution blending / solution-state rubber processing in hydrocarbon medium

**styrene content below conventional SBR levels** - St/V values include 21/63, 21/64, 21/65, 21/66 and 23/34; no explicit wt% styrene for the requested target material is disclosed

**Relevant Experimental Evidence** - Example 1: A solution blend of terminal-modified and high-molecular diene polymers was steam stripped, water squeezed, and vacuum dried to obtain diene rubber. - Example 3: Different styrene/vinyl microstructures were combined to show improved abrasion resistance and resiliency. - Example 5: Silica-filled compounds using the blended diene rubbers showed improved abrasion resistance and processability.

**Technical Relevance** This patent is relevant because it includes styrene-butadiene rubber components with explicit styrene/vinyl ratios and demonstrates how solution blending can tune processability and performance. Although not a direct synthesis patent for low-styrene SBR alone, it is important for SBR-based rubber formulation and molecular-weight balancing.

## Patent 7

- Patent Number: US12129321B2
- Patent Title: Cross-linked SBR microsphere binder and preparation method thereof and lithium- ...
- Assignee: Zhuhai Cosmx Battery Co., Ltd.
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: The patent discloses preparation of a cross-linked SBR microsphere binder by first removing polymerization inhibitors from styrene and butadiene, then carrying out aqueous emulsion polymerization with pore-forming agents, emulsifiers, redox components, and an oxidizing agent. The modified SBR latex is then demulsified, extracted, and washed to yield porous microspheres for lithium-ion battery binders. - Sodium hydroxide aqueous solution: 8% — Example 1 - Temperature: 20 °C — Example 1 - Time: 4 h — Example 1 - Deionized water: 200 parts — Example 1 - Styrene: 25 parts — Example 1 - n-Heptane: 80 parts — Example 1 - n-Propanol: 20 parts — Example 1 - Methanol: 20 parts — Example 1 - Glycerin: 40 parts — Example 1 - Xylene: 40 parts — Example 1 - Butadiene: 75 parts — Example 1 - Dodecyl mercaptan: 0.16 parts — Example 1 - Disproportionated rosin acid soap: 4.62 parts — Example 1 - Sodium methylene bisnaphthalene sulfonate: 0.15 parts — Example 1 - Sodium dithionite dihydrate: 0.025 parts — Example 1 - Potassium phosphate: 0.24 parts — Example 1 - Ferrous sulfate: 0.01 parts — Example 1 - EDTA-disodium salt: 0.01 parts — Example 1 - Cumene hydrogen peroxide: 0.06 parts — Example 1 - Pressure: 0.1 MPa — Example 1 - Polymerization temperature: 8 °C — Example 1 - Stirring speed: 100 r/min — Example 1 - Polymerization time: 7 h — Example 1 - Sodium dimethyl dithiocarbamate: 0.1 parts — Example 1 - Sodium nitrite: 0.02 parts — Example 1 - Sodium polysulfide: 0.02 parts — Example 1 - Polyvinylamine: 0.02 parts — Example 1 - Sodium chloride solution: 20% — Example 1 - Dilute sulfuric acid: 0.5% — Example 1 - Washing temperature: 50 °C — Example 1 - Diethyl ether: organic solvent extraction at 50 °C — Example 1 - Particle size: 10 nm to 100 nm — Example 1 - Porosity: 0.01% to 3% — Example 1 - Pore diameter: greater than 0 nm and less than or equal to 10 nm — Example 1 - Sodium hydroxide aqueous solution: 10% — Example 2 - Temperature: 40 °C — Example 2 - Time: 10 min — Example 2 - Deionized water: 400 parts — Example 2 - Styrene: 50 parts — Example 2 - n-Octane: 100 parts — Example 2 - Cyclohexane: 50 parts — Example 2 - Cyclohexanol: 30 parts — Example 2 - 2-Ethylbutanol: 30 parts — Example 2 - n-Butyl ether: 30 parts — Example 2 - Benzene: 50 parts — Example 2 - Acetonitrile:

10 parts — Example 2 - Butadiene: 150 parts — Example 2 - Polyethylene glycol methyl methacrylate: 10 parts — Example 2 - Polyethylene glycol diacrylate: 2 parts — Example 2 - Methyl methacrylate: 8 parts — Example 2 - Dithiodiisopropyl xanthate: 0.352 parts — Example 2 - Fatty acid soap: 10.16 parts — Example 2 - Naphthalenesulfonic acid: 0.33 parts — Example 2 - Sodium dithionite dihydrate: 0.082 parts — Example 2 - Potassium phosphate: 0.908 parts — Example 2 - Ferrous sulfate: 0.022 parts — Example 2 - EDTA-disodium salt: 0.456 parts — Example 2 - Hydrogen peroxide p-menthane: 0.24 parts — Example 2 - Sodium dimethyl dithiocarbamate: 0.22 parts — Example 2 - Sodium nitrite: 0.075 parts — Example 2 - Sodium polysulfide: 0.075 parts — Example 2 - Polyvinylamine: 0.075 parts — Example 2 - Sodium hydroxide aqueous solution: 12% — Example 3 - Temperature: 30 °C — Example 3 - Time: 2 h — Example 3 - Deionized water: 600 parts — Example 3 - Styrene: 75 parts — Example 3 - Ethyl acetate: 100 parts — Example 3 - Acetone: 50 parts — Example 3 - Diethyl phthalate: 50 parts — Example 3 - Cyclohexane: 100 parts — Example 3 - Isooctyl alcohol: 20 parts — Example 3 - n-Butyl ether: 30 parts — Example 3 - Succinonitrile: 20 parts — Example 3 - Chloroform: 20 parts — Example 3 - Urea: 5 parts — Example 3 - Polystyrene: 3 parts — Example 3 - Polyethylene glycol: 2 parts — Example 3 - Butadiene: 225 parts — Example 3 - Polyethylene glycol monoallyl ether: 6 parts — Example 3 - Polyethylene glycol methyl acrylate: 8 parts — Example 3 - Ethyl methacrylate: 8 parts — Example 3 - Acrylic acid: 10 parts — Example 3 - Dodecyl mercaptan: 0.528 parts — Example 3 - Disproportionated rosin acid soap: 15.25 parts — Example 3 - Sodium methylene bisnaphthalene sulfonate: 0.495 parts — Example 3 - Sodium dithionite dihydrate: 0.123 parts — Example 3 - Potassium phosphate: 1.362 parts — Example 3 - Ferrous sulfate: 0.033 parts — Example 3 - EDTA-disodium salt: 0.684 parts — Example 3 - Cumene hydrogen peroxide: 0.36 parts — Example 3 - Pressure: 0.25 MPa — Example 3 - Polymerization temperature: 7 °C — Example 3 - Stirring speed: 110 r/min — Example 3 - Polymerization time: 9 h — Example 3 - Sodium dimethyl dithiocarbamate: 0.33 parts — Example 3 - Sodium polysulfide: 0.13 parts — Example 3 - Polyvinylamine: 0.10 parts — Example 3 - Sodium nitrite: 0.10 parts — Example 3 - Sodium chloride solution: 24% — Example 3 - Dilute sulfuric acid: 0.5% — Example 3 - Washing temperature: 60 °C — Example 3 - Methanol extraction temperature: 65 °C — Example 3 - Particle size: 60 nm to 200 nm — Example 3 - Porosity: 8% to 12% — Example 3 - Pore diameter: greater than 0 nm and less than or equal to 60 nm — Example 3 - Sodium hydroxide aqueous solution: 14% — Example 4 - Temperature: 25 °C — Example 4 - Time: 1 h — Example 4 - Deionized water: 800 parts — Example 4 - Styrene: 100 parts — Example 4 - Fatty acid methyl ester: 150 parts — Example 4 - Toluene: 345 parts — Example 4 - Isopropanol: 50 parts — Example 4 - Di-n-amyl ether: 50 parts — Example 4 - Methyl ethyl ketone: 50 parts — Example 4 - Acetonitrile: 50 parts — Example 4 - Chloroform: 30 parts — Example 4 - Urea: 20 parts — Example 4 - Polyethylene glycol: 55 parts — Example 4 - Butadiene: 300 parts — Example 4 - Acrylonitrile: 10 parts — Example 4 - Methyl acrylate: 10 parts — Example 4 - Polyethylene glycol monomethyl ether methyl methacrylate: 20 parts — Example 4 - 2-Ethylhexyl acrylate: 10 parts — Example 4 - Dodecyl mercaptan: 0.72 parts — Example 4 - Fatty acid soap: 20.79 parts — Example 4 - Sodium methylenediphenylene sulfonate: 0.675 parts — Example 4 - Sodium dithionite dihydrate: 0.18 parts — Example 4 - Potassium phosphate: 2.03 parts — Example 4 - Ferrous sulfate: 0.045 parts — Example 4 - EDTA-disodium salt: 1.13 parts — Example 4 - Cumene hydrogen peroxide: 0.54 parts — Example 4 - Pressure: 0.3 MPa — Example 4 - Polymerization temperature: 5 °C — Example 4 - Stirring speed: 130 r/min — Example 4 -

Polymerization time: 12 h — Example 4 - Sodium dimethyl dithiocarbamate: 0.45 parts — Example 4 - Sodium nitrite: 0.2 parts — Example 4 - Sodium nitrite sodium polysulfide: 0.15 parts — Example 4 - Sodium nitrite polyvinylamine: 0.15 parts — Example 4 - Sodium chloride solution: 30% — Example 4 - Dilute sulfuric acid: 0.5% — Example 4 - Washing temperature: 55 °C — Example 4 - Acetone and benzene extraction temperature: 70 °C — Example 4 - Particle size: 80 nm to 300 nm — Example 4 - Porosity: 12% to 20% — Example 4 - Pore diameter: greater than 0 nm and less than or equal to 80 nm — Example 4 - Sodium hydroxide aqueous solution: 16% — Example 5 - Temperature: 35 °C — Example 5 - Time: 3 h — Example 5 - Deionized water: 500 parts — Example 5 - Styrene: 62.5 parts — Example 5 - Ethyl acetate: 100 parts — Example 5 - Acetone: 100 parts — Example 5 - Diethyl phthalate: 100 parts — Example 5 - n-Heptane: 50 parts — Example 5 - Dodecanol: 50 parts — Example 5 - Succinonitrile: 15 parts — Example 5 - 1,2-Dichloroethane: 10 parts — Example 5 - Amino acid: 10 parts — Example 5 - Polystyrene: 5 parts — Example 5 - Polyacrylic acid: 5 parts — Example 5 - Polyvinyl alcohol: 5 parts — Example 5 - Butadiene: 187.5 parts — Example 5 - Polyethylene glycol monomethyl ether methyl acrylate: 8 parts — Example 5 - Polyethylene glycol monoallyl ether: 8 parts — Example 5 - Acrylonitrile: 9 parts — Example 5 - Dodecyl mercaptan: 0.44 parts — Example 5 - Disproportionated rosin acid soap: 12.71 parts — Example 5 - Sodium methylene bisnaphthalene sulfonate: 0.4125 parts — Example 5 - Sodium dithionite dihydrate: 0.1025 parts — Example 5 - Potassium phosphate: 1.135 parts — Example 5 - Ferrous sulfate: 0.0275 parts — Example 5 - EDTA-disodium salt: 0.57 parts — Example 5 - Hydrogen peroxide p-menthane: 0.30 parts — Example 5 - Pressure: 0.15 MPa — Example 5 - Polymerization temperature: 7 °C — Example 5 - Stirring speed: 120 r/min — Example 5 - Polymerization time: 8 h — Example 5 - Sodium dimethyl dithiocarbamate: 0.275 parts — Example 5 - Sodium nitrite: 0.1 parts — Example 5 - Sodium polysulfide: 0.08 parts — Example 5 - Polyvinylamine: 0.1 parts — Example 5 - Sodium chloride solution: 26% — Example 5 - Dilute sulfuric acid: 0.5% — Example 5 - Washing temperature: 70 °C — Example 5 - Acetone extraction temperature: 55 °C — Example 5 - Particle size: 90 nm to 400 nm — Example 5 - Porosity: 29% to 35% — Example 5 - Pore diameter: greater than 0 nm and less than or equal to 90 nm — Example 5 - Sodium hydroxide aqueous solution: 9% — Example 6 - Temperature: 20 °C — Example 6 - Time: 2.5 h — Example 6 - Deionized water: 300 parts — Example 6 - Styrene: 37.5 parts — Example 6 - Fatty acid methyl ester: 100 parts — Example 6 - Diethyl phthalate: 200 parts — Example 6 - n-Octane: 100 parts — Example 6 - Di-n-amyl ether: 100 parts — Example 6 - Methyl ethyl ketone: 50 parts — Example 6 - Acetonitrile: 50 parts — Example 6 - 1,2-Dichloroethane: 70 parts — Example 6 - Amino acid: 10 parts — Example 6 - Polystyrene: 5 parts — Example 6 - Polyethylene glycol: 10 parts — Example 6 - Polyacrylonitrile: 5 parts — Example 6 - Butadiene: 112.5 parts — Example 6 - Polyethylene glycol methyl methacrylate: 10 parts — Example 6 - Methyl methacrylate: 2 parts — Example 6 - Acrylic acid: 3 parts — Example 6 - Dithiodiisopropyl xanthate: 0.264 parts — Example 6 - Disproportionated rosin acid soap: 7.63 parts — Example 6 - Sodium methylene bisnaphthalene sulfonate: 0.25 parts — Example 6 - Sodium dithionite dehydrate: 0.0615 parts — Example 6 - Potassium phosphate: 0.682 parts — Example 6 - Ferrous sulfate: 0.0165 parts — Example 6 - EDTA-disodium salt: 0.342 parts — Example 6 - Cumene hydrogen peroxide: 0.18 parts — Example 6 - Pressure: 0.25 MPa — Example 6 - Polymerization temperature: 6 °C — Example 6 - Stirring speed: 130 r/min — Example 6 - Polymerization time: 10 h — Example 6 - Sodium dimethyl dithiocarbamate: 0.165 parts — Example 6 - Sodium nitrite: 0.056 parts — Example 6 - Sodium polysulfide: 0.056 parts —

Example 6 - Polyvinylamine: 0.056 parts — Example 6 - Sodium chloride solution: 25% — Example 6 - Dilute sulfuric acid: 0.5% — Example 6 - Washing temperature: 65 °C — Example 6 - Ethylene oxide and methanol extraction temperature: 60 °C — Example 6 - Particle size: 90 nm to 400 nm — Example 6 - Porosity: 29% to 35% — Example 6 - Pore diameter: greater than 0 nm and less than or equal to 90 nm — Example 6 - Sodium hydroxide aqueous solution: 8% — Example 7 - Temperature: 28 °C — Example 7 - Time: 3.5 h — Example 7 - Deionized water: 400 parts — Example 7 - Styrene: 50 parts — Example 7 - Fatty acid methyl ester: 100 parts — Example 7 - Acetone: 50 parts — Example 7 - Toluene: 200 parts — Example 7 - Isoamyl alcohol: 100 parts — Example 7 - n-Butyl ether: 50 parts — Example 7 - Acetonitrile: 50 parts — Example 7 - 1,2-Dichloroethane: 30 parts — Example 7 - Amino acid: 10 parts — Example 7 - Polyethylene glycol monomethyl ether: 5 parts — Example 7 - Polyvinyl alcohol: 5 parts — Example 7 - Butadiene: 150 parts — Example 7 - Polyethylene glycol methyl methacrylate: 10 parts — Example 7 - Acrylic acid: 4 parts — Example 7 - Methyl methacrylate: 6 parts — Example 7 - Dodecyl mercaptan: 0.352 parts — Example 7 - Disproportionated rosin acid soap: 10.16 parts — Example 7 - Sodium methylene bisnaphthalene sulfonate: 0.33 parts — Example 7 - Sodium dithionite dihydrate: 0.082 parts — Example 7 - Potassium phosphate: 0.908 parts — Example 7 - Ferrous sulfate: 0.022 parts — Example 7 - EDTA-disodium salt: 0.456 parts — Example 7 - Hydrogen peroxide p-menthane: 0.24 parts — Example 7 - Pressure: 0.25 MPa — Example 7 - Polymerization temperature: 8 °C — Example 7 - Stirring speed: 110 r/min — Example 7 - Polymerization time: 9 h — Example 7 - Sodium dimethyl dithiocarbamate: 0.22 parts — Example 7 - Sodium nitrite: 0.075 parts — Example 7 - Sodium polysulfide: 0.075 parts — Example 7 - Polyvinylamine: 0.075 parts — Example 7 - Sodium chloride solution: 28% — Example 7 - Dilute sulfuric acid: 0.5% — Example 7 - Washing temperature: 70 °C — Example 7 - Ethanol extraction temperature: 65 °C — Example 7 - Particle size: 100 nm to 1 µm — Example 7 - Porosity: 20% to 29% — Example 7 - Pore diameter: greater than 0 nm and less than or equal to 100 nm — Example 7 - Medium & Water Role: Water is the core medium for emulsion polymerization and is also used extensively in inhibitor removal, demulsification, washing, and binder workup/extraction. - styrene content below conventional SBR levels: 25 parts styrene with 75 parts butadiene in Example 1; 50-100 parts styrene in later examples

**Medium & Water Role** - Water is the core medium for emulsion polymerization and is also used extensively in inhibitor removal, demulsification, washing, and binder workup/extraction. - Core reaction medium: Aqueous emulsion polymerization / latex binder synthesis

**styrene content below conventional SBR levels** - 25 parts styrene with 75 parts butadiene in Example 1; 50-100 parts styrene in later examples

**Relevant Experimental Evidence** - Example 1: A porous cross-linked SBR microsphere binder was prepared by aqueous emulsion polymerization, demulsification, extraction, and washing; the product had 10-100 nm particles and 0.01-3% porosity. - Example 3: A higher-solids formulation produced larger microspheres with 60-200 nm particle size and 8-12% porosity. - Example 7: The binder was prepared with 100 nm to 1 µm particles and 20-29% porosity, showing the tunability of the microsphere structure.

**Technical Relevance** This patent is relevant because it is an SBR-based material with explicit styrene/butadiene feed ratios and a water-based emulsion route, but its main innovation is a porous cross-linked microsphere binder for lithium-ion batteries rather than conventional tire-grade SBR. It still informs low-styrene SBR chemistry through monomer feed design and emulsion processing.

**PATENT COMPARISON TABLE**

| Patent       | Applicant               | Priority | Medium & Water Role                                                | styrene content below conventional SBR levels                                                      | Key Findings                                                                                                             |
|--------------|-------------------------|----------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| US11591428B2 | Bridgestone Corporation | 2023     | No water-related process step disclosed in the extracted evidence. | 20 to 30 percent by weight of the styrene-butadiene copolymer; examples at 26.6% and 25.7% styrene | Example 1: A tapered SBR cement was prepared in hexane and isolated by isopropanol coagulation; the product showed 26.6% |

|             |                                     |      |                                                                                                                                                     |                                                                                                                                                               |                                                                                                                          |
|-------------|-------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| US8530017B2 | Korea Kumho Petrochemical Co., Ltd. | 2013 | Water is the polymerization medium for emulsion SBR latex formation, and it is also used later for coagulation, washing, and latex handling/workup. | Styrene used at 260-280 g in Examples 1-7; Example 6 reports bound styrene content of 21.0 wt%                                                                | Example 1: Standard SBR latex was prepared in water and then blended with extension oil, coagulated, washed, and dried t |
| US9328176B2 | Korea Kumho Petrochemical Co., Ltd. | 2016 | Water is the emulsion polymerization medium, and aqueous acid/base solutions are also used in the ring-opening post-treatment of the polymer.       | Styrene used at 380 g in Example 1 and 260-380 g in comparative examples; no explicit wt% styrene for the target copolymer is given in the extracted evidence | Example 1: GMA-containing SBR was prepared in water and then ring-opened with 20% sulfuric acid to give a hydroxyl-funct |

|              |                                 |      |                                                                                                                                                                      |                                         |                                                                                                                          |
|--------------|---------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| EP2986647B1  | ARLANXEO<br>Deutschland<br>GmbH | 2017 | Water is the core emulsion polymerization medium, and it is also used later in latex blending, coagulation with steam and sulfuric acid, washing, and drying workup. | Bound styrene content: 21.0 %wt         | Example 1: Standard SBR latex was made in water, then oil-extended, coagulated with steam and sulfuric acid, washed, and |
| US12037432B2 | Lg Chem, Ltd.                   | 2024 | The polymerization itself is carried out in hydrocarbon solvent, but water is used later as hot water for solvent removal during isolation of the modified SBR.      | 27.0 wt % based on total polymer amount | Example 1: A modified SBR was prepared in n-hexane, quenched with ethanol, stabilized by condensation, and isolated by h |

|              |                                |      |                                                                                                                                                                 |                                                                                                                                  |                                                                                                                          |
|--------------|--------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| US5902856A   | Nippon Zeon Co., Ltd.          | 1999 | The rubber components are handled in solution, but water appears later during steam stripping/coagulation and water squeezing before drying.                    | St/V values include 21/63, 21/64, 21/65, 21/66 and 23/34; no explicit wt% styrene for the requested target material is disclosed | Example 1: A solution blend of terminal-modified and high-molecular diene polymers was steam stripped, water squeezed, a |
| US12129321B2 | Zhuhai Cosmx Battery Co., Ltd. | 2024 | Water is the core medium for emulsion polymerization and is also used extensively in inhibitor removal, demulsification, washing, and binder workup/extraction. | 25 parts styrene with 75 parts butadiene in Example 1; 50-100 parts styrene in later examples                                    | Example 1: A porous cross-linked SBR microsphere binder was prepared by aqueous emulsion polymerization, demulsification |

### 3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US11591428B2 and US12037432B2 are solution-polymerized SBR routes in hydrocarbon solvent, while US8530017B2, US9328176B2, EP2986647B1, and US12129321B2 are aqueous emulsion/latex systems.
- The lowest explicit styrene level in the selected set is EP2986647B1 Example 6 with 21.0 wt% bound styrene; US11591428B2 claims 20-30 wt% styrene and examples at 25.7-26.6 wt%; US12037432B2 examples are around 27.0 wt% styrene.

- US8530017B2 and US9328176B2 both use reactive or epoxy-functional monomers in emulsion SBR to improve silica affinity, whereas EP2986647B1 focuses on oil extension with vegetable-oil-based extender oils.
- US12129321B2 differs from the tire-tread-focused patents by targeting a porous cross-linked SBR microsphere binder for lithium-ion batteries, but it still uses aqueous emulsion polymerization and explicit styrene/butadiene feed ratios.
- Water is absent from the core reaction medium in the solution patents US11591428B2 and US12037432B2, but it is central to polymerization and downstream isolation in the emulsion patents, especially coagulation, washing, stripping, and extraction steps.
- Initiator systems vary: organolithium in the solution patents versus redox/peroxide systems in the emulsion patents; mercaptans are used in several emulsion systems and also in the battery-binder patent for molecular-weight control.

## 4. CONCLUSION

Across the selected primary patents, low-styrene SBR is realized through two main manufacturing families: hydrocarbon-solution polymerization with organolithium initiation and aqueous emulsion polymerization with redox/peroxide initiation. The most explicit low-styrene definitions appear in US11591428B2 (20-30 wt% styrene, 70-80 wt% butadiene) and EP2986647B1 (21.0 wt% bound styrene in Example 6), while US12037432B2 and the emulsion patents cluster around the mid-20 wt% styrene range. Several patents further tune performance by introducing reactive emulsifiers, epoxy-acrylate functionality, or oil extension, and by using coagulation/washing or solvent-removal workups to isolate the rubber. Collectively, these disclosures show that low-styrene SBR development is driven by balancing filler compatibility, wear resistance, processability, and downstream compound performance rather than by styrene reduction alone.

## 5. REFERENCES

- US11591428B2 | Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ... | Bridgestone Corporation | US | 2023 | <https://patents.google.com/patent/US11591428B2/en>
- US8530017B2 | Method for preparing styrene-butadiene copolymer using reactive emulsifier and ... | Korea Kumho Petrochemical Co., Ltd. | US | 2013 | <https://patents.google.com/patent/US8530017B2/en>
- US9328176B2 | Functional styrene-butadiene copolymer | Korea Kumho Petrochemical Co., Ltd. | US | 2016 | <https://patents.google.com/patent/US9328176B2/en>
- EP2986647B1 | Oil extended functionalized styrene-butadiene copolymer | ARLANXEO Deutschland GmbH | EP | 2017 | <https://patents.google.com/patent/EP2986647B1/en>
- US12037432B2 | Method for preparing modified and conjugated diene-based polymer and modified ... | Lg Chem, Ltd. | US | 2024 | <https://patents.google.com/patent/US12037432B2/en>

- US5902856A | Rubber composition of two diene rubbers | Nippon Zeon Co., Ltd. | US | 1999 | <https://patents.google.com/patent/US5902856A/en>
- US12129321B2 | Cross-linked SBR microsphere binder and preparation method thereof and lithium- ... | Zhuhai Cosmx Battery Co., Ltd. | US | 2024 | <https://patents.google.com/patent/US12129321B2/en>